

ADS-B, Friend or Foe: ADS-B Message Authentication for NextGen Aircraft

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Abstract—Legacy air traffic control systems have become outdated and cannot keep up with the trending increase in airliner traffic. With the goal of modernizing the system from one of manual air traffic positioning through radar and radio communication to a network of e-enabled aircraft, the Federal Aviation Administration (FAA) is shifting to the use of Automatic Dependent Surveillance-Broadcast (ADS-B) as a surveillance technology with higher accuracy and less man-in-the-loop dependence. However, ADS-B in its current implementation lacks the security measures needed to provide authenticity and integrity, making it an easy target for spoofing attacks.

This paper presents a message authentication scheme that (1) uses a Public Key Infrastructure (PKI) to verify all ADS-B signals are from FAA registered aircraft, (2) uses asymmetric cryptography to exchange a symmetric session key, and (3) uses this symmetric session key to validate data authenticity and integrity. The scheme description is followed by a discussion of its scalability and feasibility in context of the full aviation realm that it applies to, as well as the changes that would be needed in existing technologies.

Keywords—ADS-B; Broadcast authentication; Cryptography; e-enabled aircraft

I. INTRODUCTION

The demand for air travel has been steadily increasing while air traffic control systems, using World War II era technology, have remained unchanged and are becoming unable to handle the increased traffic. In order to operate under a heightened traffic load and embrace modern technology, the Federal Aviation Administration (FAA) began implementing their Next Generation Air Transportation System (NextGen) in 2012, with multiple phases to unroll through 2025 [1]. One aspect of this NextGen program that has attracted attention due to its security oversight is Automatic Dependent Surveillance-Broadcast (ADS-B). The key security issue, the use of a broadcast signal over an insecure data link with no data authentication or sender verification scheme, leaves an open door for aircraft spoofing, message modification, and message insertion attacks. This section will give an overview of ADS-B and the security issue at hand, followed by an examination of related work.

A. Automatic Dependent Surveillance-Broadcast (ADS-B)

ADS-B Out technology allows an aircraft to calculate its own position using a Global Navigation Satellite System (GNSS), such as the Global Positioning System (GPS), and then broadcast it to ground stations and other aircraft equipped with ADS-B In, or ADS-B receiving capability.

FAA Advisory Circular 20-165 gives ADS-B two data link options: universal access transceiver (UAT) or 1090 extended squitter (1090ES) [2].

UAT: UAT is on the 978 MHz frequency and allows for 144 or 272 bit packets, a larger packet size than 1090ES, transmitted once per second using ADS-B Out. ADS-B In capable aircraft can receive additional weather and flight information services on this data link due to its larger bandwidth; however, ADS-B Out on UAT is not currently permitted in class A airspace (i.e., above 18,000 ft MSL [3]) and therefore cannot be used by airliners. UAT is a viable option for general aviation because it offers additional services while not requiring replacing older transponders with ADS-B capable 1090ES transponders.

1090ES: 1090ES, aptly named, is on the 1090 MHz frequency. It allows for 112 bit packets transmitted twice per second, and is required for ADS-B Out in class A airspace due to the dependency of Traffic Collision Avoidance Systems (TCAS) on the 1090ES transponders.

Both data link options use pulsed radio frequency (RF) transmissions, like legacy radar systems, on frequencies with a set bandwidth. This means that any additional data added to the transmissions is decreasing the operational capacity of the channel, making an increase in packet size detrimental to the goal of increasing traffic density, especially for 1090ES.

In order to jam an ADS-B frequency, an attacker need only transmit with more power on the same frequency; this same concept also works for message injection or modification. With no authentication of ADS-B messages, the technology can only be used by ground stations after validation against less accurate sensor signals. Eavesdropping could also be considered a problem, but the open, broadcast nature of ADS-B is widely seen as a feature, allowing aircraft to be tracked in remote areas outside of ground station and primary radar range. The allure of ADS-B over legacy surveillance systems is due to this increased range, as well as accuracy. [4] examines the accuracy of different surveillance options, which can be seen in Fig. 1. The ability to trust ADS-B In technology would allow this increased accuracy to be used in flight applications, such as TCAS and cockpit traffic display units.

B. Related Work

Much work has been accomplished in the area of ADS-B security; however, most work is focused on air to ground communication, with little focus on air to air ADS-B In technology. Many ground stations are employing multilateration and data fusion approaches ([5], [6], [7]). Multilateration uses a

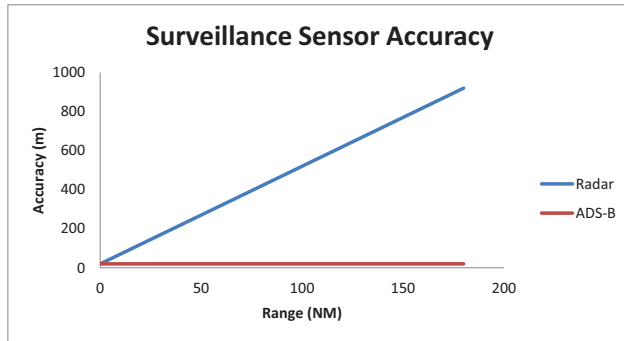


Fig. 1. Relationship between sensor accuracy (in meters) and range (in nautical miles) [4]

technique similar to modern TCAS: received signals are inherently trusted to be from an aircraft and are used to calculate the aircraft's position. In multilateration, the broadcasting aircraft's position is calculated based on the Time Difference of Arrival (TDOA) of the ADS-B signal at four different locations. This technique will not suffice in air to air communication unless there is a network of at least four trusted aircraft or ground stations within range, the capacity of which is analyzed in [8] with an open problem of establishing a trust network.

In order to establish aircraft identity, there is a need for a certificate authority for key distribution. There has been little research in the area of ADS-B cryptography due to the difficult problem of key distribution as well as the added load to the 1090 traffic frequency that would be needed for asymmetric encryption schemes. [9] proposes a symmetric cryptography scheme using a format-preserving encryption scheme, but leaves key distribution as an open problem. [10] and [11] propose symmetric key distributions using embedded secret keys in conjunction with a pilot entering a secret number or PIN with an additional interrogation scheme to thwart replay attacks.

There has also been much research on broadcast security for ad hoc networks and wireless sensor nodes that can be studied for ADS-B application. [12] suggests a method using neighborhood key exchange, but the required production of new neighborhood keys for each additional node make it infeasible for aircraft application. One interesting model for ADS-B application is Timed Efficient Stream Loss-tolerant Authentication (TESLA), which uses a symmetric key distribution in conjunction with time stamps and retroactive key publication to produce an asymmetric cryptography scheme [13]. The scheme is not scalable to aircraft, however, due to the pre-calculation of a one-way keychain which requires knowledge of flight duration time, as well as the storage overhead involved to store all aircraft ADS-B messages until a key is sent.

[14] proposes an Identity Based Encryption (IBE) scheme, which uses the aircraft's publicly known data as the public key, with the FAA acting as the certificate authority and private key generator, based on the original IBE scheme in [15]. The algorithm proposed in this paper fits into the framework of [14] but provides key details that would be needed for any real implementation. The IBE scheme will be covered in detail in Section II.

C. Contributions

This paper contributes in the following ways:

1) *Session Key Establishment*: Rather than encrypt all messages as in [14], this paper proposes to use IBE only for establishment of a session key. This session key is then used in a symmetric authentication scheme.

2) *Description of Software Updates*: This paper describes the specific avionics software updates needed for the additional ADS-B In and Out functionalities for use in this authentication scheme.

3) *Symmetric Cryptography Authentication*: A symmetric cryptography authentication scheme using SHA-3 ([16]) is described, along with discussion of the related security measures to be considered.

4) *Key Details for Implementation*: In order for this authentication scheme to be used in real applications, many key details must be considered. Some of these include: (1) description of FAA key generation algorithm, (2) time periods for individual private key updates, (3) ADS-B message size discussions, (4) private key and session key lengths, and (5) discussion of security issues and difficulties related to the authentication scheme.

II. IDENTITY BASED ENCRYPTION (IBE)

IBE can use any arbitrary string as a public key, with a private key reverse engineered by a Private Key Generator (PKG). This is useful for aerospace applications due to the expense, and in some cases inability, to access an in-flight data link for public key lookup. A preloaded database is also infeasible due to the scope of ADS-B; it may be possible for an airliner, but a hobby pilot cannot be expected to load a full database of aircraft onto his avionics every time he plans on flying in controlled airspace. [14] proposes to use a public key consisting of a concatenation of the plane's registration number, International Civil Aviation Organization (ICAO) address, tail or flight number, date, and flight information. In the original scheme by [15], IBE uses four primary functions:

Setup: This function is for the initial setup of the PKG. Given an integer security parameter and the finite message space, it returns the master public and private keys of the PKG and the global parameters needed for message decryption.

Extract: This is the private key generation task of the PKG. Given the finite message and ciphertext space, the master private key, and the public key of an individual entity, it generates a private key for that entity.

Encrypt: This is the encryption function, accomplished by any entity being interrogated for a session key. To encrypt a message, the finite message and ciphertext space, the master public key, and the individual's public key are used to produce a ciphertext.

Decrypt: This is the decryption function, accomplished by an entity that is registered with the PKG, and therefore has a private key to decrypt the received message. To decrypt a message, the finite message and ciphertext space and the individual entity's private key is used to produce the original message.

The proposed design will use IBE for in-flight aircraft identity verification as well as a means to exchange a session key to an

interrogating aircraft. Once an aircraft's session key is known, receiving aircraft can authenticate messages against a Message Authentication Code (MAC) that is appended to every ADS-B packet. Appending a MAC to ADS-B allows the broadcast to keep its open nature, while allowing other aircraft and ground stations with ADS-B In capability to authenticate both the sender and the message. The following sections will describe the design in more detail.

A. Public Key Infrastructure (PKI)

The first consideration in this design is the PKI, or in the case of IBE, the PKG. The FAA will need to act as the PKG for United States aviation. In order to globalize the scheme, there would need to be either a globally shared master private key (to the PKGs of each area), or one organization that registers aircraft globally and distributes keys to all; this could possibly be led by ICAO, with the task of key distribution disseminated to smaller PKG's based on ICAO Regional Groupings. Future work could consider scaling the proposed mechanisms to work internationally, perhaps by adding an additional layer of IBE to generate separate master private keys for each Regional Grouping in order to increase the security of such a large scale key distribution and management scheme; this would increase the practicality of the scheme, but globalization would still be challenging.

Focusing on US aviation, the FAA could develop a yearly registration process required to maintain airworthiness, possibly piggy-backing off of aircraft annual maintenance requirements. This works well with general aviation, as it allows the private key to be installed onto avionics by FAA certified personnel during an already routine procedure (e.g., annual or 100-hour inspections). [17] provides a more in-depth software and key distribution scheme for e-enabled aircraft.

Periodically, possibly bi-annually, the FAA master private and public keys will also need to be updated, which will require new key generation for all aircraft. To mitigate the impact of a sudden change, the new master public key and individual private keys should be pre-installed prior to the scheduled change date, with a set universal time to switch between these keys. Individual aircraft private keys should also be updated more frequently based on aircraft use in order to make the scheme scalable to general aviation. A privately-owned general aviation aircraft might be required to register and acquire a new private key annually, while an airliner might do so monthly. The public key of the aircraft would contain the expiration date of the key, which is not to exceed the current date. To revoke a key, the FAA would simply not give a replacement private key once the time period has expired.

For an IBE scheme to work, an aircraft public key must be obvious to other aircraft receiving its ADS-B messages; therefore, the key expiration date must be included in an ADS-B packet. Since UAT ADS-B packets are larger and more flexible, this work will conservatively focus on 1090ES in regards to ADS-B Out packets being sent and received, but some communication (specifically, the packet containing the encrypted session key) will need to be sent either across multiple 1090ES packets or via the larger UAT packet. The Identification category 1090ES packet includes a total of 75 bits of identification information. Given only an additional 32

bits for an expiration date, this information will need to be repeated in order to generate a secure public key. For a balance between security and usability, this design proposes a public key size of 256 bits.

B. Private Key Generator (PKG)

From the original work of [15], this section will describe the setup and extract functions to be used by the FAA, acting as the PKG. This private key generation scheme is based on application of the Weil pairing, first introduced by André Weil for Jacobians of curves, to an elliptic curve cryptography scheme for tripartite Diffie-Hellman key exchange [18].

1) *Setup*: The setup function will need to be run every time the FAA changes its private and public keys. It takes an input security parameter $k > 2$ in $\mathbb{Z}$ and randomly chooses a k -bit prime q . Using this q , the smallest prime p is found that satisfies: (1) $p \equiv 2 \pmod{3}$, (2) $\exists x \in \mathbb{Z} : qx = p + 1$, and (3) $q^2 \nmid p + 1$. Let E be the curve $y^2 = x^3 + 1$, and denote the projection of E over the prime field $\mathbb{F}_p$ as $E(\mathbb{F}_p)$. The $p + 1$ points over $E(\mathbb{F}_p)$ make up the cyclic group $\mathbb{G}_1$. $\mathbb{G}_2$ is the group of points over $E(\mathbb{F}_{p^2})$ that are linearly independent of the points in $E(\mathbb{F}_p)$; this is made possible by using a distortion map ϕ over the points in $\mathbb{G}_1$ based on a parameter ζ , which is a cube root of unity (i.e., solution to $x^3 = 1$, with $x \neq 1$).

In order to calculate a mapping $\hat{e} : \mathbb{G}_1 \times \mathbb{G}_1 \rightarrow \mathbb{G}_2$, a modified Weil pairing of points on the curve $E(\mathbb{F}_{p^2})$ must be used. The Weil pairing gives a mapping e such that $\forall P, Q \in \mathbb{G}_1 : \hat{e}(P, Q) = e(P, \phi(Q))$. The mapping e allows for linear independence of the groups $\mathbb{G}_1$ and $\mathbb{G}_2$, using the automorphism in $\phi(x, y) = (\zeta x, y)$, where $\phi(Q) \in E(\mathbb{F}_{p^2})$ but $\phi(Q) \notin E(\mathbb{F}_p)$. The Weil pairing between points P and Q is

$$e(P, Q) = \frac{f_P(A_Q)}{f_Q(A_P)} \quad (1)$$

where A_P and A_Q are divisors equivalent to the formal sum of points P and Q , respectively, over the curve $E(\mathbb{F}_{p^2})$, f_P is the function defined such that $f_P = qA_P$, and f_Q is the function defined such that $f_Q = qA_Q$. This pairing is efficiently computable using Miller's algorithm ([19]), as can be seen in code provided by [18].

Next, a random $s \in \mathbb{Z}_q$ is chosen; this is the PKG master private key. A generator $g \in \mathbb{G}_1$ is used to calculate the master public key: $P_{master} = g(s)$. The message space M is 152 bits (24 bit ICAO address and 128 bit session key), or $\{0, 1\}^{152}$, and the ciphertext space is $\mathbb{G}_1 \times M$. Two hash functions are selected: h_1 to go from an aircraft's public key to a parameter in $\mathbb{G}_1$ and h_2 to go from a $\mathbb{G}_2$ parameter to message space M .

The parameters to be publicly shared are: (1) initial prime q , (2) $p + 1$ points in $\mathbb{G}_1$, (3) p^2 points in $\mathbb{G}_2$, (4) map $\hat{e}$, (5) generator g , (6) FAA public key $g(s)$, (7) hash function h_1 , and (8) hash function h_2 .

2) *Extract*: To extract an aircraft's private key k_{priv} , the PKG will find the unique point (x_0, y_0) using h_1 to put the desired aircraft public key k_{pub} in $\mathbb{G}_1$ space for y_0 , then compute $x_0 \in E(\mathbb{F}_p)$ such that $x_0 \equiv (y_0^2 + 1)^{1/3} \pmod{p-1}$. k_{priv} is then calculated using s and l such that $lq = p + 1$: $k_{priv} = sl(x_0, y_0) \in \mathbb{G}_1$. Further information on the PKG

algorithms can be found in [15] and [18]. The encryption and decryption schemes will be covered in Section IID.

C. Aircraft Identity Verification

In order for an aircraft to obtain a private key, the aircraft must register with the FAA. Therefore, it can be assumed that if an aircraft can decrypt a message that has been encrypted with its public key, then the aircraft is a valid registered aircraft. In flight, the ADS-B protocol will remain unchanged aside from the addition of a MAC to the ADS-B packet. Although a public and private key scheme is already in place, this design proposes to use MACs instead of digital signatures in order to reduce packet size overhead as well as the encryption and decryption computation and verification time.

Fig. 2 illustrates the proposed verification scheme: Aircraft Alpha is sending ADS-B messages twice per second to everyone within packet range (approximately 100 NM). Upon receiving a packet from Alpha, ADS-B In capable Aircraft Bravo will send an interrogation message (which includes needed parameters from his public key), requesting Alpha's session key. Upon receiving this request, Alpha will encrypt his ICAO address and current session key using Bravo's public key along with the FAA master public key, and send this to Bravo via the UAT data link, or multiple packets on the 1090ES data link. Bravo will decrypt Alpha's session key and store it with Alpha's ICAO number to check the MACs appended to Alpha's future ADS-B messages. In order for this scheme to work, all aircraft with ADS-B Out capability must include session keys and MACs.

D. Symmetric Key Exchange

Aircraft avionics will need additional software installed to handle the symmetric key exchange. This software will need to include an autogeneration function that produces a 128 bit session key upon booting. For airliners, if the avionics are not reset between flights, there needs to be a means to regenerate the session key (e.g., a reset button). The software needs to have a means to upload data, in regards to current aircraft private key, current FAA public key, and public parameters, as well as future parameters and specific Zulu (Coordinated Universal) time for FAA public key rotation. Additional software capabilities for ADS-B Out should include the ability to strip another aircraft public key from the 1090ES messages and the encryption of the aircraft's own session key using the public parameters and the 1090ES stripped public key. A Finite State Machine (FSM) diagram for additional ADS-B Out functionality can be seen in Fig. 3.

The additional ADS-B In capabilities of the software should include the ability to send session key interrogations, the decryption of another aircraft's session key using an aircraft's private key, and the storage and cleanup of ICAO addresses and session keys. Cleanup of keys from out of range aircraft could be accomplished using an extra bit at each storage entry, checked and reset to 0 every 100 NM propagation time, and marked on memory access. ADS-B In software functionality would also include symmetric key access and message authentication using these session keys. The algorithm for symmetric cryptography authentication will be explained more in Section III. A FSM diagram of the avionics ADS-B In states can be seen in Fig. 4.

To encrypt a session key, the software will need to strip the desired aircraft identification parameters from a 1090ES packet and compile them into the aircraft's public key. An aircraft's public key will be 256 bits long and comprised of information included in the 1090ES Identification packet as well as the session key interrogation message. After compiling k_{pub}^A , of the interrogating Aircraft Alpha, Aircraft Bravo will generate a random prime $r \in \mathbb{Z}_q$ and use generator g to calculate a temporary public key $g(r)$. Bravo's appended ICAO address and session key m_{sess}^B will be used to calculate message m , as seen in Eq. 2 [15]. The multiple equalities show the properties of the bilinear mapping $\hat{e}$ for use in encryption and decryption.

$$\begin{aligned} m &= (g(r), m_{sess}^B \oplus h_2(\hat{e}(h_1(k_{pub}^A), g(s))^r)) \\ &= (g(r), m_{sess}^B \oplus h_2(\hat{e}(h_1(k_{pub}^A), g)^{sr})) \end{aligned} \quad (2)$$

Decryption of another aircraft's session key requires using the receiving aircraft's stored private key. Upon receiving $m = (m_1, m_2)$, Aircraft Alpha will decrypt as seen in Eq. 3 [15].

$$\begin{aligned} m_{sess}^B &= m_2 \oplus h_2(\hat{e}(k_{priv}^A, m_1)) \\ &= m_2 \oplus h_2(\hat{e}(h_1(k_{pub}^A)^s, g(r))) \\ &= m_2 \oplus h_2(\hat{e}(h_1(k_{pub}^A), g)^{sr}) \\ &= m_{sess}^B \oplus h_2(\hat{e}(h_1(k_{pub}^A), g)^{sr}) \oplus h_2(\hat{e}(h_1(k_{pub}^A), g)^{sr}) \end{aligned} \quad (3)$$

III. SYMMETRIC CRYPTOGRAPHY AUTHENTICATION

After receipt and storage of an aircraft's session key, the avionics must also be capable of quick access and usage of session keys to validate the ADS-B messages with their MACs. Possible lightweight MAC algorithms can be seen in [9] and [10]. A major issue with the 1090ES data link is the use of small packet sizes that already have a defined API. In order to append a MAC to an ADS-B Out packet, the defined parameters or the packet size will need to change.

In the proposed algorithm, a key-hash message authentication code (HMAC) is used, since the original message is not encrypted and a single symmetric key is used for message authentication. The cryptographic strength of the hash algorithm, key size, and hash output size are the design parameters that determine the strength of a HMAC. The proposed design uses SHA-3, which is currently under review as a new standard by National Institute of Standards and Technology (NIST).

The SHA-3 algorithm, as described in [16], takes an input string S and returns a string S' , or the MAC for the purposes of this paper. Input string S will be a concatenation of the 128-bit session key, 24-bit ICAO address, and 56-bit data portion of the received ADS-B packet. There are currently four approved SHA-3 hash algorithms, and two approved extendable-output functions (XOF's), which allow for the shortening of S' at the cost of security. The four SHA-3 algorithms, SHA3-224, SHA3-256, SHA3-384, and SHA3-512, are categorized based on the size of their output S' (224, 256, 384, and 512, respectively). The longer the output, the higher the security rating (based on brute force complexity); however, for ADS-B applications there is a need for a smaller MAC size, so an

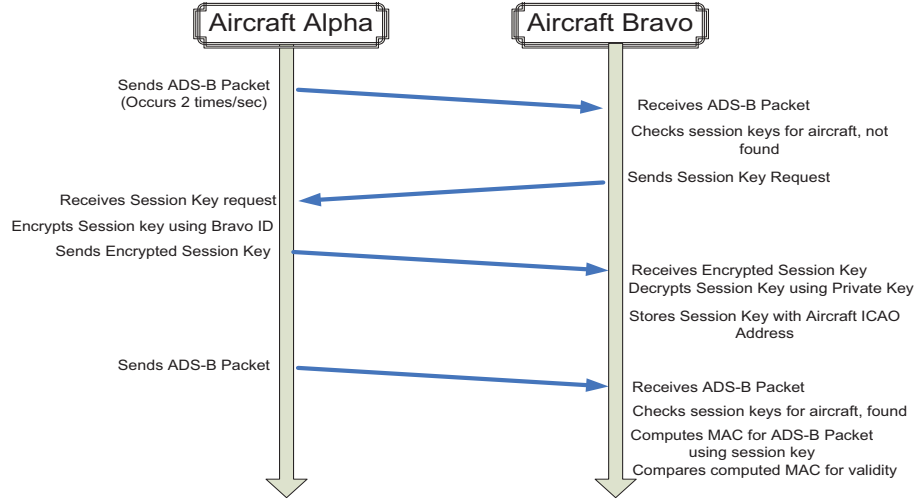


Fig. 2. Timing Diagram for ADS-B In Verification

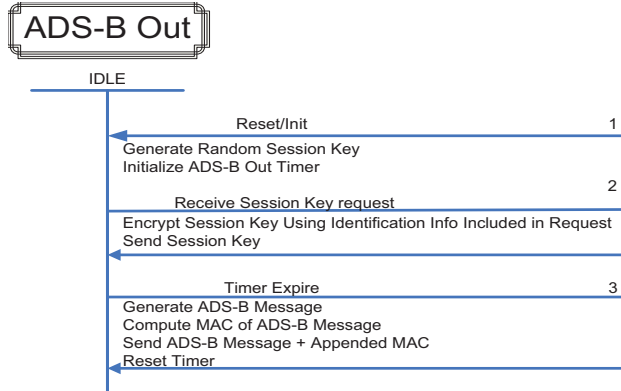


Fig. 3. ADS-B Out Finite State Machine Diagram

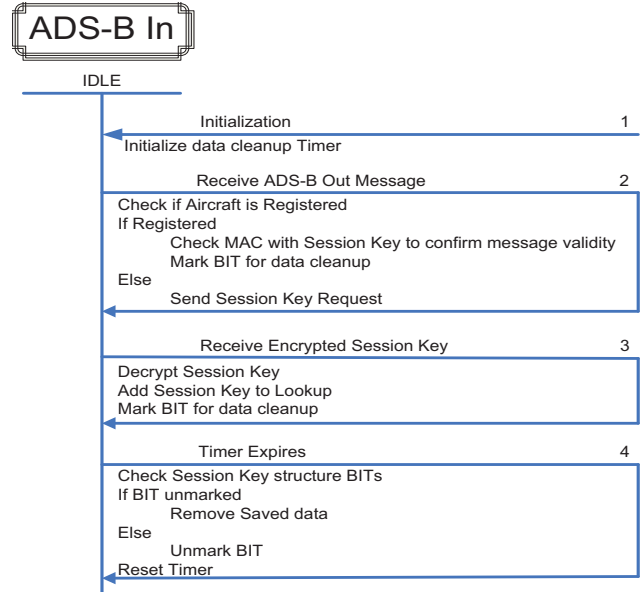


Fig. 4. ADS-B In Finite State Machine Diagram

optimum security-usability balance needs to be determined. The two XOF's, SHAKE128 and SHAKE256, are named based on their max security rating in time complexity (2^{128} and 2^{256} , respectively). The XOF's allow for user selection of output packet size; however, as packet size n decreases the time complexity rating exponentially decreases as 2^n .

The SHA-3 hash function uses: (1) a fixed length string function, (2) a positive rate integer r , and (3) a padding rule. [16] gives a number of variations of fixed length string functions, but only one is currently approved for use; this string function requires an input string of 1600 bits. This is interesting for the application of ADS-B, as the choice input string is only 208 bits, but the purpose of the padding rule is to pad the message to the correct number of bits. The functions that do not yet have approved modes of operation allow for shorter input strings, and thus will be interesting for comparison if approved. The construction of SHA-3 allows for additional design variation, but currently the approved SHA-3 hash algorithms are set with user choice only on output size, and so in turn, security.

IV. DISCUSSION AND CONCLUSIONS

This paper has presented an encryption scheme for the use of ADS-B In surveillance technology in air to air communication. An overview of the advantages and disadvantages of this scheme can be seen in Table I. An IBE scheme is used for aircraft identification and symmetric key exchange, removing the need for a costly satellite data link or a pre-loaded aircraft database, which would be inconvenient and costly for a general aviation pilot to maintain. One difficulty with this portion of the design is the added FAA workload for private key generation and the security measures needed to protect any database containing the master private key or the individual aircraft private keys. The distribution system could possibly be automated over the Internet with private keys

TABLE I. ADVANTAGES AND DISADVANTAGES OF THE PROPOSED AUTHENTICATION SCHEME

Advantages	• Provides aircraft identity verification • Authenticates ADS-B packet data • Scalable to general aviation • Uses existing data link • Avoids aircraft database public key lookup • Limits use of digital signatures to decrease overhead
Disadvantages	• Increased workload for FAA acting as PKG • Many avionics software updates needed • Increased traffic on 1090ES frequency • Difficulty of large scale key distribution and management • Single FAA master private key increases vulnerability • Appended MAC requires changes to 1090 packet size or API

downloaded from a website, a scheme that is used by many modern day certificate authorities, but this also adds more security complexities, such as proof of aircraft ownership.

Another difficult issue with the proposed IBE key exchange is the need of a new interrogation scheme across 1090ES. The interrogation could use the same packet format and size as other event driven 1090ES packets; however, the encrypted session key response will be too large for either 1090ES or UAT packets, given the 152 bit message space and 256 bit private key. This will require that the encrypted session key be split across multiple packets, adding additional complexity to the interrogation scheme, which needs to be robust to packet loss. In the current design, upon receipt of an ADS-B In packet, regardless of previous interrogations, if the session key of the broadcasting aircraft is not stored, a new interrogation will be sent. Splitting the session key across packets will require differently marked interrogation packets for each portion of the key, which is a feasible addition to the current scheme.

As for the symmetric hash authentication portion of the scheme using SHA-3, more research is needed to determine the proper balance between packet sizes and security. Using SHA-3 XOF's, a MAC can be created at any size n with a time complexity security rating of 2^n (for $n < 256$). There is no true way to determine what level of security is "secure enough," but at least for aviation purposes a brute force attack for a session key would need to be accomplished within a single session period (flight duration); in this regard, a short domestic flight would provide more security against a brute force attack than a longer international flight. Another issue is the calculation time of the HMAC; an implementation is needed to determine whether SHA-3 will suffice for this real-time application.

Looking from the point of view of general aviation pilots, this ADS-B In authentication scheme would require additional equipment to handle the computational complexities of both session key exchange and MAC calculations as well as additional storage to store aircraft session keys. This may be costly and unrealistic for hobby aircraft owners to add to older aircraft. Currently, ADS-B Out is only to be required in specific airspace, which in most cases could easily be avoided by a general aviation pilot. ADS-B In capability is not a requirement listed in the FAA's NextGen program; however, with the goal of modernizing aviation and moving towards a true network of e-enabled aircraft, the ability to trust ADS-B In communication would be a major step.

REFERENCES

- [1] Federal Aviation Administration. (2015, Feb.) NextGEN progress and plans. [Online]. Available: http://www.faa.gov/nextgen/update/progress_and_plans/
- [2] —. (2010, May) Advisory circular 20-165. [Online]. Available: http://www.faa.gov/documentLibrary/media/Advisory_Circular/AC%2020165.pdf
- [3] —. Pilot handbook. [Online]. Available: http://www.faa.gov/regulations_policies/handbooks_manuals/aviation/pilot_handbook/media/PHAK%20-%20Chapter%2014.pdf
- [4] A. Smith, R. Cassell, T. Breen, R. Hulstrom, and C. Evers, "Methods to provide system-wide ADS-B back-up, validation and security," in *25th Digital Avionics Systems Conference, 2006 IEEE/AIAA*, Oct 2006, pp. 1–7.
- [5] R. Kaune, C. Steffes, S. Rau, W. Konle, and J. Pagel, "Wide area multilateration using ADS-B transponder signals," in *Information Fusion, 2012 15th International Conference on*, July 2012, pp. 727–734.
- [6] N. Xu, R. Cassell, C. Evers, S. Hauswald, and W. Langhans, "Performance assessment of multilateration systems - a solution to nextgen surveillance," in *Integrated Communications Navigation and Surveillance Conference (ICNS), 2010*, May 2010, pp. D2–1–D2–8.
- [7] W. Liu, J. Wei, M. Liang, Y. Cao, and I. Hwang, "Multi-sensor fusion and fault detection using hybrid estimation for air traffic surveillance," *Aerospace and Electronic Systems, IEEE Transactions on*, vol. 49, no. 4, pp. 2323–2339, Oct 2013.
- [8] B. Kovell, B. Mellish, T. Newman, and O. Kajopaiye. (2012, May) Comparative analysis of ADS-B verification techniques. [Online]. Available: <http://morse.colorado.edu/~tlen5710/12s/ADSBVerification.pdf>
- [9] C. Finke, J. Butts, R. Mills, and M. Grimaila, "Enhancing the security of aircraft surveillance in the next generation air traffic control system," *International Journal of Critical Infrastructure Protection*, vol. 6, no. 1, pp. 3–11, Mar. 2013.
- [10] T.-C. Chen, "An authenticated encryption scheme for automatic dependent surveillance-broadcast data link," in *Cross Strait Quad-Regional Radio Science and Wireless Technology Conference (CSQRWC), 2012*, July 2012, pp. 127–131.
- [11] E. Valovage, "Enhanced ADS-B research," in *25th Digital Avionics Systems Conference, 2006 IEEE/AIAA*, Oct 2006, pp. 1–7.
- [12] S. Sumathy and B. Kumar, "Secure key exchange and encryption mechanism for ad hoc networks," in *Networks and Communications, 2009. NETCOM '09. First International Conference on*, Dec 2009, pp. 1–6.
- [13] A. Perrig, R. Canetti, J. D. Tygar, and D. Song. (2002) The TESLA broadcast authentication protocol. [Online]. Available: http://www.cs.berkeley.edu/~tygar/papers/TESLA_broadcast_authentication_protocol.pdf
- [14] J. Baek, Y.-J. Byon, E. Hableel, and M. Al-Qutayri, "An authentication framework for automatic dependent surveillance-broadcast based on on-line/offline identity-based signature," in *P2P, Parallel, Grid, Cloud and Internet Computing (3PGCIC), 2013 Eighth International Conference on*, Oct 2013, pp. 358–363.
- [15] D. Boneh and M. Franklin. (2001) Identity-based encryption from the weil pairing. [Online]. Available: <http://crypto.stanford.edu/~dabo/papers/ibe.pdf>
- [16] National Institute of Standards and Technology. (2014, May) SHA-3 standard: Permutation-based hash and extendable-output functions. [Online]. Available: http://csrc.nist.gov/publications/drafts/fips-202/fips_202_draft.pdf
- [17] R. V. Robinson *et al.* Secure network-enabled commercial airplane operations: IT support infrastructure challenges. [Online]. Available: <http://www.ee.washington.edu/research/nsl/papers/CEAS07.pdf>
- [18] A. E. Aftuck. (2011) The weil pairing on elliptical curves and its cryptographic applications. [Online]. Available: <http://digitalcommons.unf.edu/cgi/viewcontent.cgi?article=1138&context=etd>
- [19] V. S. Miller. (1986, May) Short programs for functions on curves. [Online]. Available: http://www.researchgate.net/publication/2551688_Short_Programs_for_functions_on_Curves